

NGUYEN THIEN THANH

AI ENGINEER

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PROFILE

Final-year AI Engineering student with hands-on experience across Generative AI, RAG pipelines, and multi-agent systems. First-authored a research paper on lightweight diffusion models (74% parameter reduction vs. baseline) and independently trained a face-verification model achieving 97.97% accuracy on 8,500 unseen identities. Comfortable owning a project end-to-end, from research and architecture to deployment.

EDUCATION

FPT University – Ho Chi Minh City

2023 – 2027 (Expected)

Bachelor of Artificial Intelligence — GPA: 3.45 / 4.0 (current)

WORK EXPERIENCE

AI Engineer | AIC Lab, FPT University

02/2026 – 5/2026

Team of 3 — Designing a RAG pipeline for financial analysis using 7B–9B parameter LLMs

- Built a query router that classifies incoming questions and directs traffic between retrieval and direct LLM responses, reducing unnecessary retrieval calls.
- Automated the data-indexing pipeline (PDF extraction → chunking → embedding) to keep the financial knowledge base continuously up to date.
- Implemented a semantic cache and duplicate-query detection layer to cut redundant LLM calls and reduce API cost and latency.
- Integrated dual input/output guardrails to enforce financial-compliance constraints on every model response.

AI Engineer Intern | Wisdom Company

05/2025 – 02/2026

Reinforcement Learning Research

- Implemented 3 core RL algorithms from scratch (Q-Learning, Policy Gradient, and MDP-based planning) in PyTorch/NumPy without relying on existing RL libraries
- Built and validated test agents against benchmark environments, then defined the team's roadmap for subsequent agent-training phases

ACADEMIC PROJECT

Research Project — First Author | “Text-to-Sketch Via Tiny-Diffusion” — FPT University

10/2025 – 12/2025

Lightweight text-to-image diffusion framework for 128×128 sketch generation, later upscaled to 512×512

- Designed a Tiny UNet (52.16M params) — a 74% reduction vs. the standard Stable Diffusion UNet (203.38M) — by fine-tuning a domain-adapted VAE and freezing it alongside CLIP during training.
- Cut training time by 51% (290 min vs. 595 min) while achieving the highest CLIPScore (25.8) among all tested models, outperforming DF-GAN (24.0), SAGAN (24.3) and the full-size Mini UNet (25.6).
- Validated results with a one-tailed t-test on 1,000 CLIPScore pairs against a GAN baseline (DF-GAN), confirming statistically significant improvement.
- Integrated a Swin2SR super-resolution stage with a custom “Canvas Isolation” technique to upscale outputs to 512×512 while eliminating edge artifacts.
- GitHub: github.com/nguyen-thanh-1/stable-diffusion-for-text-to-sketch

Face Verification Module — Face Recognition Attendance & Anti-Spoofing System

06/2026 – 08/2026

Owned end-to-end: model design, training, and evaluation of the verification module

- Trained an iResNet-50 backbone with ArcFace angular-margin loss on the MS1M-ArcFace dataset (76,504 identities, ~5M images) to produce 512-D face embeddings.
- Achieved 97.97% accuracy, 99.28% precision, and EER of 2.34% on a strictly identity-disjoint test set of 8,500 unseen identities.
- Reached 93.86% TAR at 0.1% FAR (and 89.96% at 0.01% FAR), meeting strict security thresholds for real-world deployment.
- Implemented a progressive angular-margin training schedule (0.1 → 0.5 rad) to stabilize convergence on a 82.8M-parameter model.

SKILLS

- AI & Machine Learning:** PyTorch, LLM, RAG, Diffusion Models, Transformers, Reinforcement Learning, Computer Vision, Scikit-learn
- Backend:** FastAPI, Pydantic AI (Agentic Framework), LiteLLM
- Languages & Tools:** Python (Advanced), SQL (Basic), Git/GitHub, NumPy, UV, TorchMetrics, Matplotlib
- Soft Skills:** Technical Leadership, Teamwork, Problem-solving, Self-learning

CERTIFICATES & HONORS

- Deep Learning Specialization — Andrew Ng (Coursera)
- Natural Language Processing Specialization — DeepLearning.AI (Coursera)
- Python for Everybody Specialization — University of Michigan (Coursera)
- English Proficiency: VSTEP B1 (Intermediate)
- Honorable Student of Trimester — Spring, Summer & Fall 2025
- Third Prize, University-level Science Research Competition (12/2025)